Impact of an infinite cylinder onto the free surface of a bath

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Abstract

We consider the 2-dimensional flow of an incompressible, irrotational, ideal fluid. We formulate the problem and derive simplified governing equations. We reduce the problem to the boundary using a spatial representation of the Dirichlet-to-Neumann operator which involves a singular integral, and we show the integral exists and is finite in the limit. We obtain and validate a numerical approximation of this singular integral operator by separating the integral into three different approximations. Finally, we use a finite-difference numerical scheme, namely Crank-Nicholson, to run simulations of the effects of a solid pushing on the boundary via a Gaussian pressure distribution.

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Chapter 1

Introduction

The problem of wave motion and impacts has been studied at some length, but usually only for large-scale impacts such as missile launches and seaplane landings, (Wehausen (1971), Korobkin & Pukhnachov (1988), Seddon & Moatamedi (2006)). Much of the ideas behind fluid mechanics have been studied since the early 20th century (Wagner (1932)), and as a result there is well-developed theory on impacts of rigid bodies with the surfaces of incompressible fluids (Howison et al. (1991)), but noticeably, capillary scale impacts, where the surface tension (i.e. curvature) of the fluid plays an important role (see Lee & Kim (2008)), make up a small proportion of these works (Howison et al. (2005)). In this project we consider the flow of a fluid when subjected to some external pressure, such as an infinitely long cylinder, thus formulating a 2-D problem. The impulsive motion and sinking of a cylinder at the interface has been studied by Vella and others in some depth (Vella & Li (2010), Vella et al. (2006), Greenhow (1988)), and this work hopes to add to this understanding, considering especially the dynamic version of the problem and the resulting wave following the impact.

One of the many applications of this work can be seen in Figure 1.1. Water striders are insects that are capable of floating on, and jumping off, the surface of pools of water such as ponds and lakes. This is due to their extremely hydrophobic legs, which allow them to use the surface tension of the water to prevent breaking the boundary layer of the fluid (Bush & Hu (2006)). Observe the hind legs in Figure 1.1, when magnified, resemble a cylinder which can be represented by the 2-D analysis of this work. Using the information gained from studying these fluid motions, robots have been designed with the ability to mimic the water strider by successfully "jumping" from the surface of a fluid bath (Hu et al. (2010)). The phenomena of walking

Figure 1.1: Water strider

on water is striking, and is exploited by a multitude of creatures throughout the animal kingdom, from insects and spiders to reptiles and mammals (Bush & Hu (2006)). Notable examples include the basilisk lizard and the Western Grebe, both of which have the ability to run across water due to their rapid movement creating cavities which do not collapse before they have moved on (Truscott et al. (2014)).

In general, the study of impacts on free surfaces has important uses in printing, both ink jet and 3-D (Zable (1977),Han et al. (2015)), and in agriculture. A growing global population necessitates the need for a consistent food supply in order to keep up with demand, and this is most critical in the agricultural sector. Crops are susceptible to many different threats, such as pests or diseases, each of which can decimate the potential harvest. As a result, the efficiency of pesticides is imperative, with both droplet size and the ability of pests to avoid such fluids due to their hydrophobic coverings proving to be areas of interest (Eggers & Villermaux (2008), Bush et al. (2007)). The spread of diseases in plants is also an important area of research. The proliferation of spore- and bacteria-inducing diseases among wheat crops can be commonly attributed to rainfall, see Gilet & Bourouiba (2015), and the study of the behaviour of impacting droplets, especially on plant surfaces, is vital to understand how best to minimise pathogen spread (Bourouiba et al. (2013), Wang & Bourouiba (2017), Wang & Bourouiba (2018)).

With these applications in mind, the characterisation of the cylinder impact becomes a complex free-boundary problem, and this work hopes to elucidate the behaviour of the system through analysis of the theory and computer simulation.

Chapter 2

Problem formulation

We follow Galeano-Rios et al. (2017) and Galeano-Rios et al. (2019), and references therein, and consider the two-dimensional free-surface incompressible flow of a fluid bath of uniform density ρ, subject to gravitational, interfacial and viscous forces. The domain is of infinite depth, is unbounded in the horizontal direction, and the fluid is initially at rest with its free surface undisturbed. We introduce Cartesian coordinates with the positive z axis pointing upwards. We also assume that the free surface can be described by a well-defined smooth function z = $\eta(x,t)$. Under these assumptions the flow is governed by the Navier-Stokes equations for an incompressible fluid, i.e.

$$\boldsymbol{u}_t + \boldsymbol{u}\cdot\boldsymbol{\nabla}\boldsymbol{u} = \boldsymbol{\nabla}\left(-\frac{p}{\rho}\right) + \nu\boldsymbol{\Delta}\boldsymbol{u} + \mathbf{g}, \quad z \leq \eta(x,t), \tag{2.1}$$

$$\boldsymbol{\nabla}\cdot\boldsymbol{u} = 0, \quad z \leq \eta(x,t). \tag{2.2}$$

where $\boldsymbol{u} = (u^1, u^2)^{\mathrm{T}}$ and p are the velocity and pressure field, ν is the kinematic viscosity and $\mathbf{g}$ = (0, -g) is the acceleration due to gravity. These equations are subject to the condition of no motion at infinity, i.e.

$$\mathbf{u} \to 0 \quad \text{as} \quad \sqrt{x^2+z^2} \to \infty \tag{2.3}$$

The interface is subject to surface tension and normal stresses due to the air pressure. The stress balance and kinematic condition (Johnson (1997)) require, respectively, that

$$-p\hat{\boldsymbol{n}} + \boldsymbol{\tau}\cdot\hat{\boldsymbol{n}} = (\sigma\kappa[\eta] - p_s(x,t))\hat{\boldsymbol{n}} \quad z = \eta(x,t), \tag{2.4}$$

$$\eta_t + \boldsymbol{u}\cdot\boldsymbol{\nabla}(\eta - z) = 0 \quad z = \eta(x,t). \tag{2.5}$$

Here

$$\tau = \rho\nu(\boldsymbol{\nabla u} + \boldsymbol{\nabla u^T}) = \begin{bmatrix} 2u_x^1 & u_z^1 + u_x^2 \\ u_x^2 + u_z^1 & 2u_z^2 \end{bmatrix}, \tag{2.6}$$

p_s is the pressure of the air just above the free surface,

$$\hat{\boldsymbol{n}} = \frac{1}{\sqrt{1+\eta_x^2}} \begin{bmatrix} -\eta_x \\ 1 \end{bmatrix} \tag{2.7}$$

is the outward pointing unitary normal to the free surface, σ is the surface tension coefficient and

$$\kappa[\eta] = \frac{\eta_{xx}}{(1+\eta_x^2)^{3/2}} \tag{2.8}$$

is the sum of the signed principal curvatures of the free surface, which is positive for convex functions. Notice equation (2.4) is in fact a vector equality, which yields two scalar equations. Taking the first component of this vector equation and multiplying it by $\sqrt{1+\eta_x^2}$ yields

$$p\eta_x - 2\rho\nu u_x^1\eta_x + \rho\nu u_z^1 + \rho\nu u_x^2 = -\sigma\kappa[\eta]\eta_x + p_s\eta_x. \tag{2.9}$$

We linearise around the flat surface equilibrium $\eta(x,y,t) = 0$. That is to say, we disregard non-linear terms, assuming they are negligible, and we solve the velocity field $\boldsymbol{u}$ in a fixed domain, namely the half space. Hence (2.9) becomes

$$\rho\nu u_z^1 + \rho\nu u_x^2 = 0. \tag{2.10}$$

Similarly, taking the second component gives

$$-p - \rho\nu u_x^2\eta_x - \rho\nu u_z^1\eta_x + 2\rho\nu u_z^2 = \sigma\kappa[\eta] - p_s. \tag{2.11}$$

Again we linearise around the flat surface equilibrium $\eta(x,t) = 0$, i.e. quadratic terms are considered negligible. Notice the curvature term $\kappa[\eta]$ linearises to $\partial_{xx}[\eta]$, and thus (2.11) becomes

$$-p + 2\rho\nu u_z^2 = \sigma\partial_{xx}[\eta] - p_s. \tag{2.12}$$

Recalling that $p = \tilde{p} - \rho g z$ and $\boldsymbol{g} = \boldsymbol{\nabla}(gz)$, where $\tilde{p}$ is the dynamic pressure, and incorporating conservative volume forces in the pressure term yields

$$\boldsymbol{u_t} = \boldsymbol{\nabla}\left(-\frac{\tilde{p}}{\rho}\right) + \nu\Delta\boldsymbol{u}, \qquad z \le 0, \tag{2.13a}$$

$$\boldsymbol{\nabla}\cdot\boldsymbol{u} = 0, \qquad z \le 0, \tag{2.13b}$$

$$u_z^1 + u_x^2 = 0, \qquad z = 0, \tag{2.13c}$$

$$-\tilde{p} + \rho g\eta + 2\rho\nu u_z^2 = \sigma\partial_{xx}[\eta] - p_s, \qquad z = 0, \tag{2.13d}$$

$$\eta_t = u^2, \qquad z = 0. \tag{2.13e}$$

We assume solutions satisfy the Helmholtz decomposition

$$\boldsymbol{u} = \boldsymbol{v} + \boldsymbol{w}, \tag{2.14}$$

where $\boldsymbol{v} = \boldsymbol{\nabla}\phi$ and $\boldsymbol{w} = [w^1, w^2]^T = \boldsymbol{\nabla} \times \boldsymbol{\Psi}$, with ϕ and Ψ being the scalar and vector potentials, respectively. To satisfy condition (2.3), we assume that for all $t \geq 0$,

$$\phi, \boldsymbol{\nabla}\phi \to 0 \;\text{ and }\; \Psi, \boldsymbol{w} \to 0, \quad \text{when } \sqrt{x^2 + z^2} \to \infty. \tag{2.15}$$

Substituting equation (2.14) into (2.13b) and into (2.13a) we obtain

$$\Delta\phi = 0, \quad z \leq 0, \tag{2.16}$$

$$\boldsymbol{\nabla}(\phi_t) + \boldsymbol{w_t} = \boldsymbol{\nabla}\left(-\frac{\tilde{p}}{\rho}\right) + \nu\Delta\boldsymbol{w}, \quad z \leq 0, \tag{2.17}$$

respectively. We note that (2.17) is satisfied if we choose

$$\phi_t = -\frac{\tilde{p}}{\rho}, \quad z \leq 0, \tag{2.18}$$

$$w_t^i = \nu\Delta w^i, \quad i = 1, 2;\; z \leq 0. \tag{2.19}$$

Moreover, substituting equation (2.14) into (2.13c) and (2.13d) yields

$$0 = (\phi_x + w^1)_z + (\phi_z + w^2)_x, \quad z = 0, \tag{2.20}$$

$$-\frac{\tilde{p}}{\rho} + g\eta = \frac{\sigma}{\rho}\partial_{xx}[\eta] - 2\nu(\phi_z + w^2)_z - \frac{p_s}{\rho}, \quad z = 0. \tag{2.21}$$

We take the x derivative of (2.20), and use (2.19) for i = 2 to obtain

$$w_t^2 = 2\nu\partial_{xx}(\phi_z + w^2), \quad z = 0, \tag{2.22}$$

where we used $w_x^1 + w_z^2 = 0$. We now combine equations (2.18), (2.16) and (2.21) to yield

$$\phi_t = -g\eta + \frac{\sigma}{\rho}\partial_{xx}[\eta] + 2\nu\phi_{xx} - 2\nu w_z^2 - \frac{p_s}{\rho}, \quad z = 0. \tag{2.23}$$

We take ρ as the characteristic density, define the characteristic length L, velocity V and introduce the dimensionless numbers

$$Fr = V^2/(gL), \;\; We = \rho V^2 L/\sigma \;\text{ and } Re = LV/\nu, \tag{2.24}$$

i.e. the square Froude number, Weber number and Reynolds number respectively. Then from (2.16), (2.19) with i = 2, (2.13e) and (2.14), (2.23), (2.22) and condition (2.15), we obtain a closed system which is given in dimensionless form by

$$\Delta\phi = 0, \quad z \leq 0, \tag{2.25a}$$

$$w_t^2 = \frac{1}{Re}\Delta w^2, \quad z \leq 0. \tag{2.25b}$$

$$\eta_t = \phi_z + w^2, \quad z = 0, \tag{2.25c}$$

$$\phi_t = -\frac{1}{Fr}\eta + \frac{1}{We}\partial[\eta] + \frac{2}{Re}\phi_{xx} - \frac{2}{Re}w_z^2 - p_s, \quad z = 0, \tag{2.25d}$$

$$w_t^2 = \frac{2}{Re}\partial_{xx}(\phi_z + w^2), \quad z = 0, \tag{2.25e}$$

subject to

$$\phi, \nabla\phi \to 0 \ \text{ and } \ w^2 \to 0, \quad \text{when } \ \sqrt{x^2 + z^2} \to \infty. \tag{2.26}$$

We consider cases in which the Reynolds number can attain high values. For example, in the case of water, $\nu \approx 10^{-4} cm^2/s$, so if $L = 1mm$ and $V = 30cm/s$ then

$$Re = \frac{LV}{\nu} = \frac{(0.1cm)(30cm/s)}{10^{-4}cm^2/s} = 3 \times 10^4. \tag{2.27}$$

Thus, we approximate the problem by taking the Reynolds number to be infinity. Equations (2.25) simplify to

$$\Delta\phi = 0, \quad z \leq 0, \tag{2.28a}$$

$$\eta_t = \phi_z, \quad z = 0, \tag{2.28b}$$

$$\phi_t = -\frac{1}{Fr}\eta + \frac{1}{We}\partial_{xx}[\eta] - p_s, \quad z = 0. \tag{2.28c}$$

We notice that equations (2.28b) and (2.28c) are only defined on the surface, suggesting we can reduce the problem to the interface. The only term in these two equations that requires information from inside the domain is ϕ_z in (2.28b). Therefore we find a way to express ϕ in terms of ϕ_z. We shall do this in the next chapter, deriving the Dirichlet-to-Neumann operator N(ϕ), thus reducing the 2-dimensional problem to a 1-dimensional system, since only (2.28a) is a 2-dimensional equation.

Chapter 3

Reducing the system to the boundary

Instead of solving the Laplace problem in the infinite half plane, we want to solve the problem only on the boundary, i.e. the real line. To do this, we notice that (2.28b) and (2.28c) are only on the boundary ($z = 0$), and the only term in these that directly involves information from the interior of the fluid is the ϕ_z term. Therefore, if we can find an operator that, given ϕ along the boundary, can return the value of ϕ_z along the boundary, we can reduce the problem further. This operator is precisely the Dirichlet-to-Neumann (DtN) operator for the Laplace problem in the half-space. More generally, a Dirichlet-to-Neumann operator is defined as follows: given Dirichlet data on the boundary of a domain in which Laplace's equation has a unique solution, one can solve Laplace's equation for the interior of the region, and, if the problem has a well-defined normal derivative on the boundary, use this to find the Neumann data on the boundary. The resulting map from the Dirichlet data to the Neumann data is the Dirichlet-to-Neumann operator.

In the case of impacts it is necessary to find a physical space representation of the DtN because the contact area between the cylinder and the free boundary varies continuously with time, so the domain has to be decomposed into two parts; the free boundary away from the impact, and the surface in contact with the cylinder, where η is known, but the pressure is not. Such representation is derived in the following section and it involves a singular integral operator on the boundary, which we will show exists and is finite in the limit.

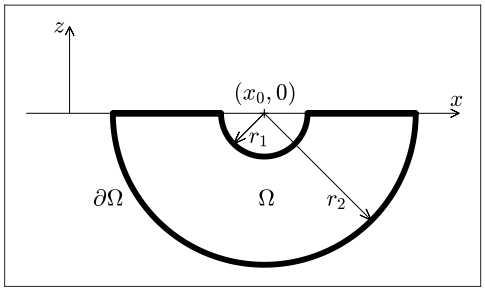


Figure 3.1: The considered domain Ω

3.1 A singular integral representation of the DtN operator in physical space

We construct a half-annulus Ω centred at an arbitrary point x_0 on the surface, with boundary $\partial\Omega$ defined in the fluid by semi-circular curves C_1 and C_2 (with radii r_1 and r_2 respectively), and on the surface by the straight lines between $x = r_1$ and $x = r_2$, and $x = -r_1$ and $x = -r_2$, (see Figure 3.1). From Green's Second Identity we have

$$\underbrace{\int_{\Omega} \psi \Delta \phi_z}_{=\,0} - \underbrace{\int_{\Omega} \phi_z \Delta \psi}_{=\,0} = \int_{\partial\Omega} \psi \partial_n \phi_z - \int_{\partial\Omega} \phi_z \partial_n \psi \tag{3.1a}$$

$$\Rightarrow \quad \int_{\partial\Omega} \psi \partial_n \phi_z = \int_{\partial\Omega} \phi_z \partial_n \psi, \tag{3.1b}$$

and the annulus construction allows us to use the general solution to Laplace's equation, given by $\psi(x, z) = -\frac{1}{2\pi} \log |r|$, where $r = |\sqrt{(x - x_0)^2 + z^2}|$, since otherwise there would be a singularity at the point x_0. The LHS of (3.1b) is equivalent to

$$\int_{C_1} \psi \underbrace{\partial_n}_{=-\partial_r} \phi_z + \int_{C_2} \psi \underbrace{\partial_n}_{=\partial_r} \phi_z + \int_{-r_2}^{-r_1} \psi \underbrace{\partial_n}_{=\partial_z} \phi_z + \int_{r_1}^{r_2} \psi \underbrace{\partial_n}_{=\partial_z} \phi_z \tag{3.2}$$

and the RHS equivalent to

$$\int_{C_1} \phi_z \underbrace{\partial_n}_{=-\partial_r} \psi + \int_{C_2} \phi_z \underbrace{\partial_n}_{=\partial_r} \psi + \underbrace{\int_{-r_2}^{-r_1} \phi_z \underbrace{\partial_n}_{=\partial_z} \psi}_{=0} + \underbrace{\int_{r_1}^{r_2} \phi_z \underbrace{\partial_n}_{=\partial_z} \psi}_{=0}, \tag{3.3}$$

where the normal derivative of ψ along $z = 0$ is zero, since that boundary is perpendicular to its gradient. Equation (3.1b) then becomes

$$\underbrace{-\int\limits_{C_1} \psi\partial_r\phi_z}_{I_1'} + \underbrace{\int\limits_{C_2} \psi\partial_r\phi_z}_{I_2'} + \underbrace{\int\limits_{-r_2}^{-r_1} \psi\phi_{zz}}_{I_3'} + \underbrace{\int\limits_{r_1}^{r_2} \psi\phi_{zz}}_{I_4'} = \int\limits_{C_2} \phi_z\partial_r\psi - \int\limits_{C_1} \phi_z\partial_r\psi. \quad (3.4)$$

Taking the derivative of ψ with respect to r yields

$$\partial_r\psi = -\frac{1}{2\pi r}, \quad (3.5)$$

so the RHS of (3.4) becomes

$$\underbrace{\frac{1}{2\pi r_1}\int\limits_{C_1} \phi_z}_{J_1} + \underbrace{-\frac{1}{2\pi r_2}\int\limits_{C_2} \phi_z}_{J_2}. \quad (3.6)$$

Furthermore, $\phi_{zz} = -\phi_{xx}$ (as ϕ is Laplacian free), and since $\psi = -\frac{1}{2\pi}\log|r|$ is independent of θ in the integrand, the LHS of (3.4) can be written as

$$I_1' = \underbrace{\frac{1}{2\pi}\log|r_1|\int\limits_{C_1} \partial_r\phi_z}_{I_1}, \quad (3.7a)$$

$$I_2' = \underbrace{\Big(-\frac{1}{2\pi}\log|r_2|\Big)\int\limits_{C_2} \partial_r\phi_z}_{I_2}, \quad (3.7b)$$

$$I_3' = \underbrace{\frac{1}{2\pi}\int\limits_{-r_2}^{-r_1} \log|u|\phi_{xx}\mathrm{d}u}_{I_3}, \quad (3.7c)$$

$$I_4' = \underbrace{\frac{1}{2\pi}\int\limits_{r_1}^{r_2} \log|u|\phi_{xx}\mathrm{d}u}_{I_4}, \quad (3.7d)$$

where $u = x - x_0$. That is, $I_1 + I_2 + I_3 + I_4 = J_1 + J_2$, and we assume $\phi_z \to 0$ sufficiently fast when $(x, z) \to \infty$, and $J_2 \to 0$ and $I_2 \to 0$ when $r_2 \to \infty$, i.e. taking the limit of $r_2 \to \infty$, we have $I_1 + I_3 + I_4 = J_1$, where the lower limit

in I_3 and the upper limit in I_4 are now taken to be ∞.
Now, rewriting J_1 in terms of u and Taylor expanding, we have

$$J_1 = \frac{1}{2\pi r_1} \int\limits_{C_1} \phi_z(x_0, 0) + \epsilon(x - x_0, z) \tag{3.8a}$$

$$= \frac{\phi_z(x,0)}{2\cancel{\pi r_1}} \cdot \underbrace{\cancel{\pi r_1}}_{\text{length of arc}} + \underbrace{\frac{1}{2\pi r_1} \int\limits_{C_1} \epsilon(x - x_0, z)}_{J_{1a}} \tag{3.8b}$$

where, by Taylor's Theorem, $|\epsilon| \leq \delta\sqrt{(x-x_0)^2 + z^2}$ for some $\delta > 0$. Hence

$$|J_1 a| \leq \frac{1}{2\pi r_1} \cdot \delta \int\limits_{C_1} \sqrt{(x-x_0)^2 + z^2} = \frac{1}{2\cancel{\pi r_1}} \delta r_1 \cancel{\pi r_1} = \frac{\delta r_1}{2} \xrightarrow{r_1 \to 0} 0. \tag{3.9}$$

So $J_1 \to \frac{\phi_z}{2}$.
Turning our attention to I_1, we observe

$$|I_1| \leq \frac{1}{2\pi} |\log(r_1)| \int\limits_{C_1} \sup_{r_1 < \rho} |\partial_n \phi| \quad \forall r_1 \leq \rho \tag{3.10a}$$

$$\leq \frac{1}{2\pi} |\log(r_1)| \sup_{r_1 < \rho} |\partial_n \phi| \pi r_1 \tag{3.10b}$$

$$\to 0 \text{ as } r_1 \to 0; \tag{3.10c}$$

since $\lim_{x \to 0}(x \ln x) = 0$. The remaining terms I_3 and I_4 can be written as

$$I_3 = \frac{1}{2\pi} \int\limits_{-\infty}^{-r_1} \psi(x - x_0, 0) \phi_{xx}(x - x_0, 0) \tag{3.11}$$

and

$$I_4 = \frac{1}{2\pi} \int\limits_{r_1}^{\infty} \psi(x - x_0, 0) \phi_{xx}(x - x_0, 0), \tag{3.12}$$

and integration by parts yields

$$I_3 = \frac{1}{2\pi} \Bigg(- \int\limits_{-\infty}^{-r_1} \frac{\phi(u,0)}{u^2} \mathrm{d}u + \frac{\phi(x_0 - r_1, 0)}{r_1} - \lim_{B \to \infty} \left(\frac{\phi(x_0 - B, 0)}{B} \right)$$
$$+ \underbrace{\phi(x_0 - r_1, 0) \log(r_1)}_{I_{3a}} - \lim_{B \to \infty} \left(\phi_x(x_0 - B, 0) \log(B) \right) \Bigg) \tag{3.13}$$

and

$$I_4 = \frac{1}{2\pi}\left(-\int_{r_1}^{\infty}\frac{\phi(u,0)}{u^2}\mathrm{d}u + \frac{\phi(x_0+r_1,0)}{r_1} - \lim_{B\to\infty}\left(\frac{\phi(x_0+B,0)}{B}\right)\right.$$

$$\left.\underbrace{-\phi_x(x_0+r_1,0)\log(r_1)}_{I_{4a}} + \lim_{B\to\infty}\left(\phi_x(x_0+B,0)\log(B)\right)\right). \quad (3.14)$$

We assume $\phi, \nabla\phi$ and its derivatives decay to zero sufficiently fast when $\pm r_2 \to \pm\infty$; hence the terms involving r_2 tend to zero in the limit. Taking said limit, the limits of integration of I_3 and I_4 become$\int_{-\infty}^{x_0-r_1}$ and $\int_{x_0+r_1}^{\infty}$ respectively. Moreover, combining I_{3a} and I_{4a} and taking the limit $r_1 \to 0$ gives

$$\log(r_1)\Big[\phi_x(x_0-r_1,0) - \phi_x(x_0+r_1,0)\Big] \quad (3.15a)$$

$$\Leftrightarrow -\log(r_1)\Big[\phi_x(x_0+r_1,0) - \phi_x(x_0-r_1,0)\Big] \quad (3.15b)$$

$$\Leftrightarrow \underbrace{-2r_1\log(r_1)}_{\to\ 0 \text{ as } r_1\ \to\ 0}\underbrace{\frac{\Big[\phi_x(x_0+r_1,0) - \phi_x(x_0-r_1,0)\Big]}{2r_1}}_{\to\ \phi_{xx}(x_0,0) \text{ as } r_1\ \to\ 0}, \quad (3.15c)$$

so these terms reduce to zero.
This leaves

$$I_3 = \frac{1}{2\pi}\left(-\int_{-\infty}^{x_0-r_1}\frac{\phi(u,0)}{u^2}\mathrm{d}u + \frac{\phi(x_0-r_1,0)}{r_1}\right) \quad (3.16)$$

and

$$I_4 = \frac{1}{2\pi}\left(-\int_{x_0+r_1}^{\infty}\frac{\phi(u,0)}{u^2}\mathrm{d}u + \frac{\phi(x_0+r_1,0)}{r_1}\right). \quad (3.17)$$

Further inspection of I_3 and I_4 shows that

$$\frac{\phi(x_0-r_1,0)}{r_1} = \frac{\phi(x_0,0)}{r_1} - \frac{\phi_x r_1}{r_1} + \frac{\epsilon(x_0-r_1)}{r_1}, \quad (3.18)$$

where $\frac{|\epsilon(x_0-r_1)|}{r_1} \to 0$ as $r_1 \to 0$, and

$$\frac{\phi(x_0+r_1,0)}{r_1} = \frac{\phi(x_0,0)}{r_1} + \frac{\phi_x r_1}{r_1} + \frac{\epsilon(x_0+r_1)}{r_1}, \quad (3.19)$$

where $\frac{|\epsilon(x_0+r_1)|}{r_1} \to 0$ as $r_1 \to 0$.
Noticing

$$\frac{1}{r_1} = \int_{r_1}^{\infty} \frac{1}{x^2}, \tag{3.20}$$

taking the limit of $r_2 \to \infty$ and recalling $u = x - x_0$ we obtain

$$\frac{1}{r_1} = \int_{x_0+r_1}^{\infty} \frac{1}{(x-x_0)^2} \quad \text{and} \quad \frac{1}{r_1} = \int_{-\infty}^{-r_1} \frac{1}{(x-x_0)^2} \tag{3.21}$$

$$\implies \frac{\phi(x_0,0)}{r_1} = \int_{x_0+r_1}^{\infty} \frac{\phi(x_0,0)}{(x-x_0)^2} \quad \text{and} \quad \frac{\phi(x_0,0)}{r_1} = \int_{-\infty}^{x_0-r_1} \frac{\phi(x_0,0)}{(x-x_0)^2}. \tag{3.22}$$

Therefore

$$I_3 = \frac{1}{2\pi}\left(-\int_{-\infty}^{x_0-r_1} \frac{\phi(x,0)}{(x-x_0)^2} + \int_{-\infty}^{x_0-r_1} \frac{\phi(x_0,0)}{(x-x_0)^2} \right) + \text{zero terms} \tag{3.23}$$

$$= \frac{1}{2\pi}\left(\int_{-\infty}^{x_0-r_1} \frac{\phi(x_0,0)-\phi(x,0)}{(x-x_0)^2} \right), \tag{3.24}$$

and

$$I_4 = \frac{1}{2\pi}\left(-\int_{x_0+r_1}^{\infty} \frac{\phi(x,0)}{(x-x_0)^2} + \int_{x_0+r_1}^{\infty} \frac{\phi(x_0,0)}{(x-x_0)^2} \right) + \text{zero terms} \tag{3.25}$$

$$= \frac{1}{2\pi}\left(\int_{x_0+r_1}^{\infty} \frac{\phi(x_0,0)-\phi(x,0)}{(x-x_0)^2} \right). \tag{3.26}$$

Thus, in the limit, we have

$$\frac{\phi_z(x_0,0)}{\cancel{2}} = \frac{1}{\cancel{2}\pi}\left(\lim_{\epsilon\to 0} \int_{\mathbb{R}\setminus(x_0-\epsilon,x_0+\epsilon)} \frac{\phi(x_0,0)-\phi(x,0)}{(x-x_0)^2} \right), \tag{3.27}$$

which gives the general formula for ϕ_z as

$$\phi_z(x_0,0) = \frac{1}{\pi}\left(\lim_{\epsilon\to 0} \int_{\mathbb{R}\setminus(x_0-\epsilon,x_0+\epsilon)} \frac{\phi(x_0,0)-\phi(x,0)}{(x-x_0)^2} \right). \tag{3.28}$$

3.1.1 Convergence of the singular integral operator

Notice that equation (3.28) above defines an operator (N) which takes a function ϕ defined on the real line which is smooth and bounded, and yields a function $N(\phi)$. We have shown in the section before that if ϕ is the Dirichlet condition for a Laplace problem in the half-space ($z \leq 0$), then $\phi_z = N(\phi)$. However, we still need to show that N is well defined, i.e. the singular integral limit converges at every point x_0.

We emphasise that at each point x_0 on the boundary, we calculate ϕ_z as the limit of a function of ϵ which is given by an integral containing a singularity at x_0, that is, as $\epsilon \to 0$, the integrand becomes infinitely large. To show that the limits exists, we notice that away from the singularity it is enough that ϕ remains bounded. So, outside of the interval $(x_0 - M, x_0 + M)$, for some constant M, the integral can be computed, either exactly or by numerical approximation. Therefore the problem remains only in the vicinity of the singularity, in the interval $(x_0 - M, x_0 + M) \setminus (x_0 - \epsilon, x_0 + \epsilon)$.

Now, Taylor expanding $\phi(x, 0)$ gives

$$\phi(x,0) = \phi(x_0,0) + \phi_x(x_0,0)(x - x_0) + \frac{1}{2}\phi_{xx}(x_0,0)(x - x_0)^2 + \text{h.o.t.} \quad (3.29)$$

and substituting into (3.28) yields

$$\phi_z(x_0,0) = \frac{1}{\pi}\left[\lim_{\epsilon\to 0}\left(\int\limits_{(x_0-M,x_0+M)\setminus(x_0-\epsilon,x_0+\epsilon)} \cancel{\frac{\phi(x_0,0)}{(x-x_0)^2}} - \cancel{\frac{\phi(x_0,0)}{(x-x_0)^2}} - \frac{\phi_x(x_0,0)\cancel{(x-x_0)}}{(x-x_0)^{\cancel{2}}} - \frac{\phi_{xx}(x_0,0)\cancel{(x-x_0)^2}}{2\cancel{(x-x_0)^2}} - \frac{\epsilon}{(x_0-x)^2}\right)\right] \quad (3.30)$$

$$\Rightarrow \phi_z(x_0,0) = \frac{1}{\pi}\left[\lim_{\epsilon\to 0}\left(\int\limits_{(x_0-M,x_0+M)\setminus(x_0-\epsilon,x_0+\epsilon)} -\frac{\phi_x(x_0,0)}{(x-x_0)} - \underbrace{\frac{\phi_{xx}(x_0,0)}{2}}_{\text{finite}} - \underbrace{\frac{\epsilon}{(x_0-x)^2}}_{\text{finite}}\right)\right]. \quad (3.31)$$

Since $\phi_x(x_0, 0)$ is constant, and $(x - x_0)^{-1}$ is an odd function, the first term of (3.31) is antisymmetric about the singularity, so it cancels for each $\epsilon > 0$,

yielding a zero in the limit. The second term is a constant function which integrates to a finite value in the limit, and the error term converges to zero by Taylor's Theorem, so the final term of (3.31) must be bounded by some constant. Hence the limit exists and is finite.

Chapter 4

Numerical approximation of the Dirichlet-to-Neumann operator

In order to run simulations in which we find numerical solutions to the free surface elevation when it is pushed on by an impacting object, we need to find a numerical approximation of the singular integral operator given by (3.28). We notice that the DtN operator is linear in the argument function, and therefore we can construct a matrix operator that approximates it in a discretised domain. This spatial representation of the operator allows for a natural treatment of the decomposition of the domain into the pressed area and the rest of the surface, which is free to move.

4.1 Three different approximations

Recall the general formula for the Dirichlet-to-Neumann operator (3.28)

$$\phi_z(x_0, 0) = \frac{1}{\pi}\left(\lim_{\epsilon\to 0} \int\limits_{\mathbb{R}\setminus(x_0-\epsilon, x_0+\epsilon)} \frac{\phi(x_0, 0) - \phi(x, 0)}{(x - x_0)^2} \mathrm{d}x\right), \tag{4.1}$$

which (suppressing the z argument) can be written as

$$\phi_z(x_0) = \underbrace{\frac{1}{\pi}\lim_{\epsilon\to 0}\int\limits_{(x_0-M,x_0+M)\setminus(x_0-\epsilon,x_0+\epsilon)} \frac{\phi(x_0)-\phi(x)}{(x-x_0)^2}\mathrm{d}x}_{\mathrm{L}}$$
$$+\underbrace{\frac{\phi(x_0)}{\pi}\int\limits_{(-\infty,x_0-M)\cup(x_0+M,\infty)} \frac{1}{(x-x_0)^2}\mathrm{d}x}_{P_1}$$
$$+\underbrace{\frac{1}{\pi}\int\limits_{(-\infty,x_0-M)\cup(x_0+M,\infty)} \frac{-\phi(x)}{(x-x_0)^2}\mathrm{d}x}_{P_2}. \quad (4.2)$$

4.1.1 Away from the singularity: Exact integration of P_1

Evaluating P_1, we have

$$P_1 = \frac{\phi(x_0)}{\pi}\left(\int\limits_{-\infty}^{x_0-M}\frac{\mathrm{d}x}{(x-x_0)^2} + \int\limits_{x_0+M}^{\infty}\frac{\mathrm{d}x}{(x-x_0)^2}\right), \quad (4.3)$$

and using $x = x_0 - M \implies x - x_0 = M$,

$$P_1 = \frac{\phi(x)}{\pi}\left(\int\limits_{-\infty}^{-M}\frac{\mathrm{d}(x-x_0)}{(x-x_0)^2} + \int\limits_{M}^{\infty}\frac{\mathrm{d}(x-x_0)}{(x-x_0)^2}\right) \quad (4.4)$$
$$= \frac{\phi(x_0)}{\pi}\left(\left[-\frac{1}{x}\right]_{-\infty}^{-M} + \left[-\frac{1}{x}\right]_{M}^{\infty}\right) \quad (4.5)$$
$$= \frac{\phi(x_0)}{\pi}\left(\frac{1}{M}+\frac{1}{M}\right) \quad (4.6)$$
$$= \frac{2\phi(x_0)}{\pi M} \quad (4.7)$$
$$= \frac{\phi(x_0)}{\pi\Delta_x}, \quad (4.8)$$

where $M = 2\Delta_x$ with Δ_x being the size of the spatial grid consisting of N_x intervals.

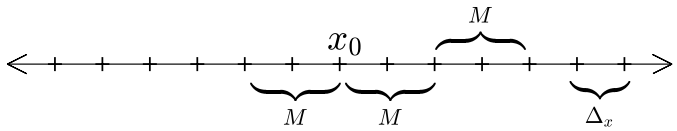


Figure 4.1: Spatial mesh for the problem and its separation into the regions around, and away from, the singularity.

4.1.2 Away from the singularity: Simpson's Rule for P_2

Recall P_2 is given by

$$P_2 = \frac{1}{\pi} \int\limits_{(-\infty, x_0 - M) \cup (x_0 + M, \infty)} \frac{-\phi(x)}{(x - x_0)^2} \mathrm{d}x, \tag{4.9}$$

and consider the positive half of the region of integration, that is,

$$P_3 \coloneqq \frac{1}{\pi} \int\limits_{x_0 + M}^{B} \frac{-\phi(x)}{(x - x_0)^2} \mathrm{d}x, \tag{4.10}$$

where we assume B is a constant large enough such that waves have decayed beyond B, so that the integral (4.10) becomes

$$P_3 = \sum_{n \in Y} -\frac{1}{\pi} \int\limits_{x_0 + (n-1)\Delta_x}^{x_0 + (n+1)\Delta_x} \frac{\phi(x)}{(x - x_0)^2} \mathrm{d}x, \tag{4.11}$$

where $Y = \{n + i; i = 3, 5, 7, \ldots, N_x + 1 - n_0\}$ and $x_0 = \Delta_x n_0$. Notice that each term in the sum (4.11) corresponds to the integration in an interval of length $2\Delta_x$, which is centred at a point $x_n = x_0 + n\Delta_x$. Within each of these intervals, we wish to approximate $\phi(x)$ by a second order polynomial $P(s) = a_0 + a_1 s + a_2 s^2$, where $s = \frac{x - x_0 - n\Delta_x}{\Delta_x}$; where $s(x = n\Delta_x) = 0$, $s(x = (n-1)\Delta_x) = -1$ and $s(x = (n+1)\Delta_x) = 1$. We denote the value of $P(s)$ at $s = 0, 1$ and -1 as ϕ_0, ϕ_1 and ϕ_{-1} respectively. Therefore, we have

$$P(0) = a_0 = \phi_0, \tag{4.12a}$$

$$P(1) = \phi_0 + a_1 + a_2 = \phi_1, \tag{4.12b}$$

$$P(-1) = \phi_0 - a_1 + a_2 = \phi_{-1}, \tag{4.12c}$$

so

$$P(1) = a_1 + a_2 = \phi_1 - \phi_0, \tag{4.13}$$

$$P(-1) = -a_1 + a_2 = \phi_{-1} - \phi_0, \tag{4.14}$$

adding (4.13) and (4.14),

$$\Rightarrow 2a_2 = \phi_1 - 2\phi_0 + \phi_{-1}, \tag{4.15}$$

$$\Rightarrow a_2 = \frac{\phi_1 - 2\phi_0 + \phi_{-1}}{2}. \tag{4.16}$$

Substituting into (4.13) gives

$$2a_1 = \phi_1 - \phi_{-1} \tag{4.17}$$

$$\Rightarrow a_1 = \frac{\phi_1 - \phi_{-1}}{2}, \tag{4.18}$$

i.e.

$$a_0 = \phi_0, \quad a_1 = \frac{\phi_1 - \phi_{-1}}{2}, \quad a_2 = \frac{\phi_{-1} - 2\phi_0 + \phi_1}{2}. \tag{4.19}$$

Changing variables, we have

$$P_3 = \sum_{n\in Y} -\frac{\cancel{\Delta_x}}{\pi} \int_{-1}^{1} \frac{P(s)}{\Delta_x^{\cancel{2}}(s+n)^2}\,\mathrm{d}s = \sum_{n\in Y} -\frac{1}{\pi\Delta_x} \int_{-1}^{1} \frac{P(s)}{(s+n)^2}\,\mathrm{d}s. \tag{4.20}$$

Hence

$$P_3 = \sum_{n\in Y} -\frac{1}{\pi\Delta_x} \int_{-1}^{1} \frac{a_0 + a_1 s + a_2 s^2}{(s+n)^2}\,\mathrm{d}s \tag{4.21}$$

introducing $w = s + n$,

$$= \sum_{n\in Y} -\frac{1}{\pi\Delta_x} \int_{n-1}^{n+1} \frac{a_0 + a_1(w-n) + a_2(w-n)^2}{w^2}\,\mathrm{d}w \tag{4.22}$$

$$= \sum_{n\in Y} -\frac{1}{\pi\Delta_x} \int_{n-1}^{n+1} \underbrace{\frac{a_0}{w^2}}_{(i)} + \underbrace{\frac{a_1(w-n)}{w^2}}_{(ii)} + \underbrace{\frac{a_2(w-n)^2}{w^2}}_{(iii)}\,\mathrm{d}w. \tag{4.23}$$

Evaluating each term of the integrand individually $\left(\text{remembering the factor of } -\frac{1}{\pi\Delta_x}\right)$ gives

$$(i) = -\frac{a_0}{\pi\Delta_x} \int_{n-1}^{n+1} \frac{\mathrm{d}w}{w^2} = \frac{a_0}{\pi\Delta_x}\left[\frac{1}{w}\right]_{n-1}^{n+1} = \frac{a_0}{\pi\Delta_x}\left(\frac{1}{n+1} - \frac{1}{n-1}\right), \tag{4.24}$$

$$(ii) = -\frac{a_1}{\pi\Delta_x}\left[\int_{n-1}^{n+1}\frac{\cancel{w}}{w^{\cancel{2}}}\mathrm{d}w - n\int_{n-1}^{n+1}\frac{\mathrm{d}w}{w^2}\right] \tag{4.25}$$

$$= -\frac{a_1}{\pi\Delta_x}\left(\left[\ln w\right]_{n-1}^{n+1} + n\left[\frac{1}{w}\right]_{n-1}^{n+1}\right) \tag{4.26}$$

$$= -\frac{a_1}{\pi\Delta_x}\left(\ln\left(\frac{n+1}{n-1}\right) - \frac{2n}{(n+1)(n-1)}\right), \tag{4.27}$$

$$(iii) = -\frac{a_2}{\pi\Delta_x}\int_{n-1}^{n+1}\frac{w^2+n^2-2nw}{w^2}\mathrm{d}w \tag{4.28}$$

$$= -\frac{a_2}{\pi\Delta_x}\left(\int_{n-1}^{n+1}\mathrm{d}w + \int_{n-1}^{n+1}\frac{n^2}{w^2} - \int_{n-1}^{n+1}\frac{2n}{w}\mathrm{d}w\right) \tag{4.29}$$

$$= -\frac{a_2}{\pi\Delta_x}\left(\left[w\right]_{n-1}^{n+1} - n^2\left[\frac{1}{w}\right]_{n-1}^{n+1} - 2n\left[\ln w\right]_{n-1}^{n+1}\right) \tag{4.30}$$

$$= -\frac{a_2}{\pi\Delta_x}\left(2 + \frac{2n^2}{(n+1)(n-1)} - 2n\ln\left(\frac{n+1}{n-1}\right)\right). \tag{4.31}$$

So, combining (4.19), (4.24), (4.27) and (4.31) yields

$$\begin{aligned} P_3 = \sum_{n\in Y}\frac{1}{\pi\Delta_x}\Bigg[&\phi_0\left(4 - 2n\ln\left(\frac{n+1}{n-1}\right)\right) \\ &+ \phi_1\left(-\frac{n}{(n+1)} + \left(n-\frac{1}{2}\right)\ln\left(\frac{n+1}{n-1}\right) - 1\right) \\ &+ \phi_{-1}\left(-\frac{n}{(n-1)} + \left(n+\frac{1}{2}\right)\ln\left(\frac{n+1}{n-1}\right) - 1\right)\Bigg]. \end{aligned} \tag{4.32}$$

This allows us to calculate every linear weight of the right-hand half of the nodes of our mesh. By symmetry and translational invariance of the problem this is all we need, as we can simply find the weights on the left-hand side by reflections. Notice that the sum (4.32) can be taken as going to infinity, since $\phi_k = 0\ \forall\ k \geq N_x + 1, \forall\ k \leq 0$.

4.1.3 In the vicinity of the singularity: Evaluating L

Finally, consider L as in (4.2)

$$\mathrm{L} = \frac{1}{\pi} \lim_{\epsilon \to 0} \int\limits_{(x_0 - M, x_0 + M) \setminus (x_0 - \epsilon, x_0 + \epsilon)} \frac{\phi(x_0) - \phi(x)}{(x - x_0)^2} \mathrm{d}x. \tag{4.33}$$

We approximate L using a fourth order polynomial, $Q(m) = b_4 m^4 + b_3 m^3 + b_2 m^2 + b_1 m + b_0$, with $m = \frac{x - n_0 \Delta_x}{\Delta_x}$, and hence

$$m(x_0) = 0 \tag{4.34a}$$
$$m(x_0 + \Delta_x) = 1 \tag{4.34b}$$
$$m(x_0 + 2\Delta_x) = 2 \tag{4.34c}$$
$$m(x_0 - \Delta_x) = -1 \tag{4.34d}$$
$$m(x_0 - 2\Delta_x) = -2; \tag{4.34e}$$

notice that in particular $b_0 = 0$, as $Q(m = 0) = \phi(x_0) - \phi(x_0)$.

Again, using the discretised domain (Figure 4.1), with the singularity at the point $x_0 = n_0 \Delta_x$, and substituting the polynomial approximation in the numerator of (4.33) gives

$$\mathrm{L} \approx \frac{1}{\pi} \lim_{\epsilon \to 0} \int\limits_{(n_0 \Delta_x - 2\Delta_x, n_0 \Delta_x + 2\Delta_x) \setminus (n_0 \Delta_x - \epsilon, n_0 \Delta_x + \epsilon)} \frac{Q(\frac{x - n_0 \Delta_x}{\Delta_x})}{(x - n_0 \Delta_x)^2} \mathrm{d}x, \tag{4.35}$$

changing the variable to m and defining $\upsilon \coloneqq \epsilon / \Delta_x$, we obtain

$$\mathrm{L} \approx \frac{1}{\Delta_x \pi} \lim_{\upsilon \to 0} \int\limits_{(-2,2) \setminus (-\upsilon, \upsilon)} \frac{Q(m)}{m^2} \, \mathrm{d}m; \tag{4.36}$$

i.e.

$$\mathrm{L} \approx \frac{1}{\Delta_x \pi} \lim_{\upsilon \to 0} \int\limits_{(-2,2) \setminus (-\upsilon, \upsilon)} \frac{b_1}{m} + b_2 + b_3 m + b_4 m^2 \, \mathrm{d}m; \tag{4.37}$$

Note that the term b_1/m of (4.37) is zero as $1/m$ is an odd function, so it is anti-symmetric about the origin and the integral cancels for all $\upsilon > 0$, and therefore the limit of the integral of that term is zero. We also observe that the term $b_3 m$ is an odd function which is also anti-symmetric about 0, and therefore cancels out. This leaves the only the even powers, i.e.

$$\mathrm{L} \approx \frac{1}{\Delta_x \pi} \int_{-2}^{2} b_2 + b_4 m^2 \mathrm{d}m. \tag{4.38}$$

Thus, recovering the original variables we have

$$L \approx \frac{4b_2}{\pi \Delta_x} + \frac{16 b_4}{3\pi\Delta_x}. \tag{4.39}$$

Using $u = \frac{x - n_0\Delta_x}{\Delta_x}$, we have

$$Q(0) = b_0 = 0, \tag{4.40a}$$
$$Q(1) = b_4 + b_3 + b_2 + b_1 = \phi(n_0\Delta_x) - \phi(n_0\Delta_x + \Delta_x), \tag{4.40b}$$
$$Q(2) = 16b_4 + 8b_3 + 4b_2 + 2b_1 = \phi(n_0\Delta_x) - \phi(n_0\Delta_x + 2\Delta_x), \tag{4.40c}$$
$$Q(-1) = b_4 - b_3 + b_2 - b_1 = \phi(n_0\Delta_x) - \phi(n_0\Delta_x - \Delta_x), \tag{4.40d}$$
$$Q(-2) = 16b_4 - 8b_3 + 4b_2 - 2b_1 = \phi(n_0\Delta_x) - \phi(n_0\Delta_x - 2\Delta_x). \tag{4.40e}$$

In matrix form, this is

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 16 & 8 & 4 & 2 \\ 1 & -1 & 1 & -1 \\ 16 & -8 & 4 & -2 \end{bmatrix} \begin{bmatrix} b_4 \\ b_3 \\ b_2 \\ b_1 \end{bmatrix} = \begin{bmatrix} \phi(n_0\Delta_x) - \phi(n_0\Delta_x + \Delta_x) \\ \phi(n_0\Delta_x) - \phi(n_0\Delta_x + 2\Delta_x) \\ \phi(n_0\Delta_x) - \phi(n_0\Delta_x - \Delta_x) \\ \phi(n_0\Delta_x) - \phi(n_0\Delta_x - 2\Delta_x) \end{bmatrix}, \tag{4.41}$$

so inverting the matrix on the left hand side gives the b_i's. Since we only require the b_2 and b_4, (4.41) yields

$$b_2 = \frac{1}{24}\left[\phi((n_0-2)\Delta_x) - 16\phi((n_0-1)\Delta_x) + 30\phi(n_0\Delta_x) \right. \\ \left. -16\phi((n_0+1)\Delta_x) + \phi((n_0+2)\Delta_x)\right], \tag{4.42}$$

and

$$b_4 = \frac{1}{24}\left[-\phi((n_0-2)\Delta_x) + 4\phi((n_0-1)\Delta_x) - 6\phi(n_0\Delta_x) \right. \\ \left. +4\phi((n_0+1)\Delta_x) - \phi((n_0+2)\Delta_x)\right]. \tag{4.43}$$

Substituting (4.42) and (4.43) into (4.39), we have

$$\frac{4}{\pi\Delta_x}\left(b_2 + \frac{4}{3}b_4\right) = \frac{1}{6\pi\Delta_x}\left[\phi_{-2} - 16\phi_{-1} + 30\phi_0 - 16\phi_1 + \phi_2 \right. \\ \left. + \frac{4}{3}\left[-\phi_{-2} + 4\phi_{-1} - 6\phi_0 + 4\phi_1 - \phi_2\right]\right] \tag{4.44}$$

$$\Rightarrow \frac{4}{\pi\Delta_x}\left(b_2+\frac{4}{3}b_4\right)=\frac{1}{6\pi\Delta_x}\left[-\frac{1}{3}\phi_{-2}-\frac{32}{3}\phi_{-1}+22\phi_0-\frac{32}{3}\phi_1-\frac{1}{3}\phi_2\right], \tag{4.45}$$

which gives the approximation for L.

4.1.4 The approximation of ϕ_z

In order to combine the approximations for L, P_1 and P_2, we observe that the point x_0 is simply $n_0\Delta_x$ intervals from the left hand side of the mesh, so the local weighting ϕ_i in (4.45) becomes $\phi((n_0+i)\Delta_x)$, whilst in (4.32), the ϕ_i become $\phi((n_0+n+i)\Delta_x)$ for the positive half of P_2, and $\phi((n_0-n-i)\Delta_x)$ for the negative half. Therefore, (4.8), (4.32) and (4.45) give the numerical approximation for ϕ_z to be

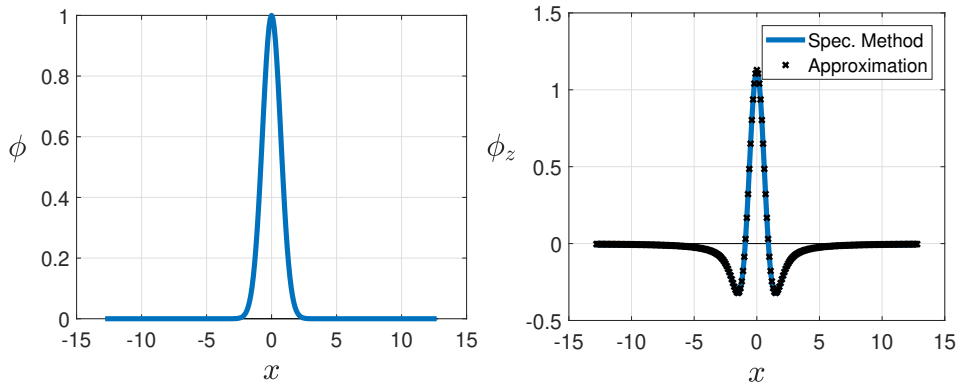
$$\begin{aligned}\phi_z(x_0)\approx\frac{1}{6\pi\Delta_x}&\left[-\frac{1}{3}\phi((n_0-2)\Delta_x)-\frac{32}{3}\phi((n_0-1)\Delta_x)+22\phi(n_0\Delta_x)\right.\\&\left.-\frac{32}{3}\phi((n_0+1)\Delta_x)-\frac{1}{3}\phi((n_0+2)\Delta_x)\right]+\frac{\phi(x_0)}{\pi\Delta_x}\\+\sum_{n\in Y}\frac{1}{\pi\Delta_x}&\left[f_{-1}(n)\phi((n_0+n-1)\Delta_x)+f_0(n)\phi((n_0+n)\Delta_x)+f_1(n)\phi((n_0+n+1)\Delta_x)\right]\\+\sum_{n\in Y}\frac{1}{\pi\Delta_x}&\left[f_{-1}(n)\phi((n_0-n+1)\Delta_x)+f_0(n)\phi((n_0-n)\Delta_x)+f_1(n)\phi((n_0-n-1)\Delta_x)\right],\end{aligned} \tag{4.46}$$

where $f_{-1}(n)=\left(-\frac{n}{(n-1)}+\left(n+\frac{1}{2}\right)\ln\left(\frac{n+1}{n-1}\right)-1\right)$, $f_0(n)=\left(4-2n\ln\left(\frac{n+1}{n-1}\right)\right)$ and $f_1(n)=\left(-\frac{n}{(n+1)}+\left(n-\frac{1}{2}\right)\ln\left(\frac{n+1}{n-1}\right)-1\right)$. See Appendix A for the code to construct the numerical approximation of the N operator.

4.2 Validation of the method

In order to validate the proposed numerical method (4.46), we compare the output of our method to that of a well established one, a spectral method. Using a Gaussian curve as the initial data, Figure 4.2a, we compare the results for ϕ_z using both methods (see Figure 4.2b). Table 4.1 shows that the average absolute error between our approximation and a spectral method decreases as the number of mesh points, N_x, increases, demonstrating that the error decays as we increase the mesh refinement.

As will be seen in the following chapter, the reason we do not use a spectral method is that in order to simulate a solid pressing against the free



(a) Gaussian data

(b) Comparison

Figure 4.2: Left: Gaussian data. Right: Comparison between the numerical approximation to the singular integral operator and the spectral method to find the DtN transform of the Gaussian data on the left.

surface we need to decompose the physical domain into two regions; that is, we need only replace the free-surface elevation at the contact points with the height of the impacting object. This decomposition is possible with our DtN operator N, as the free-surface elevation is given by entries of a matrix (see Chapter 5), so replacing the elevation at the contact points simply consists of altering the columns of said matrix.

We highlight that the structure of the matrix reflects the translational invariance in the problem; the weights loaded in the DtN matrix are simply shifts of the previously loaded rows (Figure 4.3). This is because the values of the Dirichlet boundary condition in the immediate vicinity of a point have the greatest effect on the Neumann boundary condition, whereas the values further away from a point provide a lesser contribution, due to waves decaying towards infinity. It is this invariance under translation that reduces the calculations required for the approximation of P_2, as only the positive

Mesh Points	Average Error
32	0.009532840385648
64	0.001771519459574
128	0.001412301494288
256	0.001412258158996

Table 4.1: Average Absolute Errors

half of the domain needs to be considered, with the negative half obtained by reflections.

Thus, our method is a valid approximation of ϕ_z over a discretised domain, and hence we can begin to run simulations of different pressures on the boundary; the next chapter hopes to elucidate this process using a Gaussian pressure distribution.

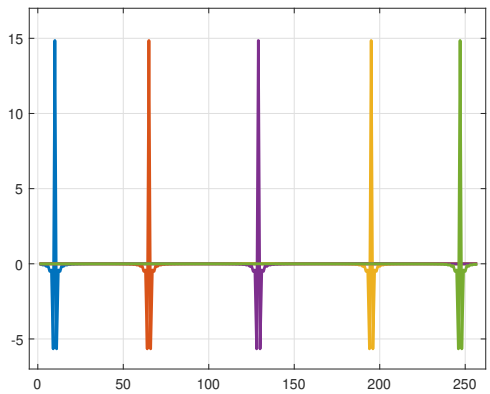


Figure 4.3: Various rows of N.
x axis: Row Index
y axis: Element Value

See Appendix A for the code used to create Figures (4.2a), (4.2b) and (4.3).

Chapter 5

Simulations

Following the validation of method (4.46), we wish to simulate varying pressures on the free boundary. Recall the 2-dimensional system of equations (2.28) as derived in Chapter 2,

$$\Delta\phi = 0, \quad z \leq 0, \tag{5.1a}$$

$$\eta_t = \phi_z, \quad z = 0, \tag{5.1b}$$

$$\phi_t = -\frac{1}{Fr}\eta + \frac{1}{We}\partial_{xx}[\eta] - p_s, \quad z = 0, \tag{5.1c}$$

which, introducing the DtN operator N, becomes the 1-dimensional problem

$$\eta_t = N(\phi), \tag{5.2a}$$

$$\phi_t = -\frac{1}{Fr}\eta + \frac{1}{We}\partial_{xx}[\eta] - p_s. \tag{5.2b}$$

In order to apply the approximation to (5.2) we use the following numerical scheme.

5.1 The Crank-Nicholson Scheme

The Crank-Nicholson scheme (Strikwerda (2004)) replaces finite derivatives with finite differences, by taking the average of the previous and current time on the RHS of the discretised system of equations. Thus the temporal derivative in (5.2a) is approximated by

$$\frac{\hat{\eta}_i^{k+1} - \hat{\eta}_i^k}{\Delta_t} = \left(\frac{\boldsymbol{N}}{2}\varphi^{k+1}\right)_i + \left(\frac{\boldsymbol{N}}{2}\varphi^k\right)_i, \tag{5.3}$$

where the superscript is the temporal step, the subscript is the spatial step, $\hat{\eta}$ and φ are the numerical approximations of η and ϕ, and $\boldsymbol{N}(\phi^k) \approx \phi_z^k(n\Delta_x)$. Rearranging so that all terms involving the $k+1$ step are on the LHS, we have

$$\hat{\eta}_i^{k+1} - \frac{\Delta_t}{2}\left(\boldsymbol{N}\varphi^{k+1}\right)_i = \hat{\eta}_i^k + \frac{\Delta_t}{2}\left(\boldsymbol{N}\varphi^k\right)_i. \tag{5.4}$$

Similarly, (5.2b) is given by

$$\frac{\varphi_i^{k+1} - \varphi_i^k}{\Delta_t} = -\left(\frac{\hat{\eta}_i^{k+1} + \hat{\eta}_i^k}{2Fr}\right) + \frac{\left(\partial_{xx}\hat{\eta}^{k+1}\right)_i}{2We} + \frac{\left(\partial_{xx}\hat{\eta}^k\right)_i}{2We} - \frac{\left(p_s^k\right)_i}{2} - \frac{\left(p_s^{k+1}\right)_i}{2}, \tag{5.5}$$

where $\partial_{xx}\hat{\eta}^{k+1} = \frac{\hat{\eta}_{i-1}^{k+1} - 2\hat{\eta}_i^{k+1} + \hat{\eta}_{i+1}^{k+1}}{\Delta_x^2}$. Thus, rearranging for the $k+1$ terms (but leaving the pressure terms together) yields

$$\begin{aligned}\varphi_i^{k+1} + \frac{\Delta_t}{2Fr}\hat{\eta}_i^{k+1} - \frac{\Delta_t}{2We}\left(\frac{\hat{\eta}_{i-1}^{k+1} - 2\hat{\eta}_i^{k+1} + \hat{\eta}_{i+1}^{k+1}}{\Delta_x^2}\right) = \\ -\frac{\Delta_t}{2Fr}\hat{\eta}_i^k + \frac{\Delta_t}{2We}\left(\frac{\hat{\eta}_{i-1}^k - 2\hat{\eta}_i^k + \hat{\eta}_{i+1}^k}{\Delta_x^2}\right) - \frac{\Delta_t}{2}\left((p_s^{k+1})_i + (p_s^k)_i\right) + \varphi_i^k.\end{aligned} \tag{5.6}$$

Setting $\beta = \frac{\Delta_t}{2Fr}$ and $\gamma = \frac{\Delta_t}{2We\Delta_x^2}$, we have

$$\begin{aligned}\varphi_i^{k+1} - \gamma\hat{\eta}_{i-1}^{k+1} + (\beta + 2\gamma)\,\hat{\eta}_i^{k+1} - \gamma\hat{\eta}_{i+1}^{k+1} = \\ \gamma\hat{\eta}_{i-1}^k - (\beta + 2\gamma)\,\hat{\eta}_i^k + \gamma\hat{\eta}_{i+1}^k - \frac{\Delta_t}{2}\left((p_s^{k+1})_i + (p_s^k)_i\right) + \varphi_i^k.\end{aligned} \tag{5.7}$$

Combining (5.4) and (5.7) in matrix form, we have

$$\begin{bmatrix} I & \alpha\boldsymbol{N} \\ \boldsymbol{A} & I \end{bmatrix}\begin{bmatrix}\hat{\eta} \\ \varphi\end{bmatrix}^{k+1} = \begin{bmatrix} I & -\alpha\boldsymbol{N} \\ -\boldsymbol{A} & I \end{bmatrix}\begin{bmatrix}\hat{\eta} \\ \varphi\end{bmatrix}^{k} + \alpha\begin{bmatrix} 0 \\ p^{k+1} + p^k \end{bmatrix}, \tag{5.8}$$

where $\alpha = -\Delta_t/2$, and $\boldsymbol{A}$ has the form

$$\begin{bmatrix} (\beta+2\gamma) & -\gamma & 0 & \cdots & 0 \\ -\gamma & (\beta+2\gamma) & -\gamma & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -\gamma & (\beta+2\gamma) & -\gamma \\ 0 & \cdots & 0 & -\gamma & (\beta+2\gamma) \end{bmatrix}. \tag{5.9}$$

5.2 Simulation of a perturbed free surface

Using the Crank-Nicholson scheme in the form (5.8), we can simulate the perturbed free surface for different values of p_s. As can be seen in `Simulation.m` in Appendix A, we choose a Gaussian pressure distribution (Figure 5.1) given by

$$\left(\frac{1-\cos\left(\frac{2\pi}{\tau}t\right)}{2}\right)e^{-\left(\frac{x}{l}\right)^2}. \tag{5.10}$$

where we apply the pressure over the interval $t \in [0, \tau]$. Our choice of τ and l determine the dimensionless Froude and Weber numbers. Recall, from (2.24) we have

$$Fr = V^2/(gL), \quad We = \rho V^2 L/\sigma, \tag{5.11}$$

$$\Rightarrow Fr = L/T^2 g, \quad We = \rho L^3/T^2\sigma, \tag{5.12}$$

as $V^2 = L^2/T^2$. Thus, taking $L = l$, $T = \tau$ and $M = \rho L^3$, the pressure distribution is nondimensionalised to

$$\left(\frac{1-\cos\left(2\pi\hat{t}\right)}{2}\right)e^{-(\hat{x})^2}. \tag{5.13}$$

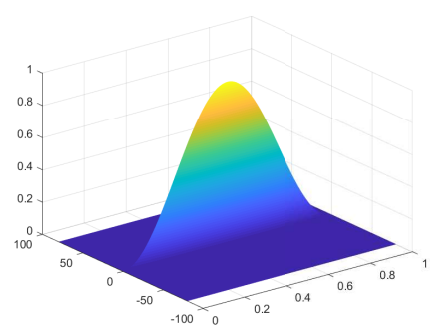
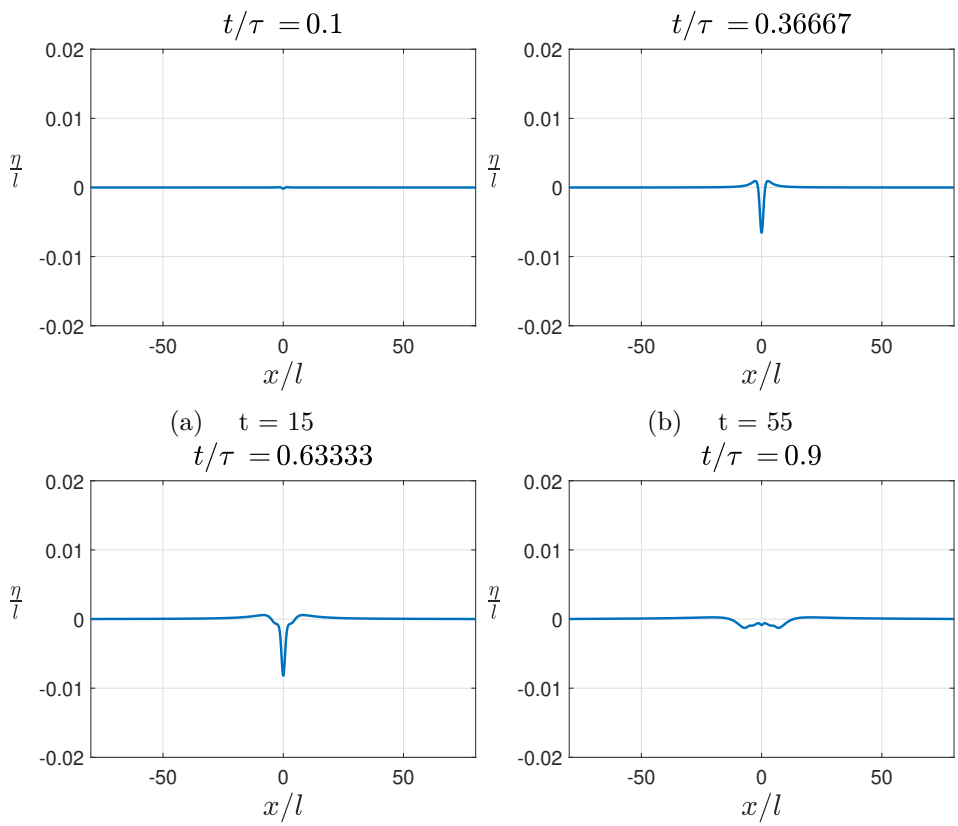


Figure 5.1: The Gaussian pressure distribution

The Gaussian pressure distribution (5.13) represents a pressure being applied to the surface and then withdrawn, similar to the behaviour of an impacting object. Implementing (5.13) in `Simulation.m`, Figure 5.2 shows the movement of the wave caused by this pressure distribution as time progresses.



(a) t = 15 (b) t = 55

(c) t = 95 (d) t = 135

Figure 5.2: Surface profile as the wave moves at different timesteps

Chapter 6

Discussion

From the governing equations of an incompressible, irrotational ideal fluid we formulated the problem and reduced it to the boundary using the Dirichlet-to-Neumann operator. We derived a singular integral representation of the DtN operator and separated the region of integration in order to obtain a numerical approximation, which we validated and used to conduct simulations of pressure applied to the boundary.

Our model is an accurate approximation to the problem, and appears to give realistic simulations of a perturbed free surface. We can easily adjust the pressure term in the code `Simulation.m` to simulate the impact of a cylinder onto the surface, allowing for a range of scenarios to be visualised. Furthermore, the model could be improved by coupling the motion of an impacting object with that of the free boundary, in order to simulate both the impact and rebound of an object.

Since we are working at capillary scale, the affect of curvature κ on the system can also be included after the linearisation stage, to further reduce the error of the approximation, but this is not necessary for an accurate impact simulation, so is not discussed here.

These ideas can be extended to higher dimensions, namely 3-D, in which the cylinder can be replaced by a sphere, and more advanced simulations can be achieved (Galeano-Rios et al. (2019)). These axisymmetric realisations (Galeano-Rios et al. (2017)) use the same basic principles as are discussed in this work, but applied to a more complex problem, and are the next step in this field of research.

Appendix A

Computer code

DtNmatrix.m

```
close all
clear
clc

dx = .1;
m = 2^8;
line = zeros(1,m+1);
line(1) = 1+11/3;
line(2) = -16/9;
line(3) = -1/18;

for nn = 3:2:m
    an = -nn/(nn-1) +(nn+1/2)*log((nn+1)/(nn-1)) -1; % f_-1(n)
    bn =                -2*nn*log((nn+1)/(nn-1)) +4; % f_0(n)
    cn = -nn/(nn+1) +(nn-1/2)*log((nn+1)/(nn-1)) -1; % f_1(n)
    line(nn) = line(nn)+an;
    line(nn+1) = line(nn+1)+bn;
    line(nn+2) = line(nn+2)+cn;
end
line = line/(pi*dx);

N = zeros(m+1,m+1); % Construction of N
for ii = 1:m+1
    N(ii,:) = [fliplr(line(2:ii)),line(1:m-ii+2)];
end

x = dx*(-m/2:m/2)';
phi=exp(-x.^2);
phiz = N*phi;
```

```
figure  % plot of the gaussian data
plot(x,zeros(size(x)),'k')
hold on
phiSpec = phi(1:end-1);
dkx = 2*pi/(m*dx);
kx = [0:m/2 -m/2+1:-1]'*dkx;
phizSpec = real(ifft(abs(kx).*fft(phiSpec)));
plot(x(1:end-1),phiSpec,'LineWidth',4)
grid on
set(gca,'FontSize',16)
xlabel('$x$','interpreter','latex','FontSize',24)
ylabel('$\phi\␣\␣\␣\␣$','interpreter','latex','FontSize',24,'
    rotation',0)
print('-depsc','Phi.eps')

figure  % plot of the comparison
plot(x,zeros(size(x)),'k')
hold on
a = plot(x(1:end-1),phizSpec,'LineWidth',4);
b = plot(x,phiz,'xk','MarkerSize',6,'LineWidth',2);
set(gca,'FontSize',16)
legend([a,b],'Spec.␣Method','Approximation')
grid on
set(gca,'FontSize',16)
xlabel('$x$','interpreter','latex','FontSize',24)
ylabel('$\phi_z\␣\␣\␣\␣$','interpreter','latex','FontSize',24,'
    rotation',0)
print('-depsc','PhiZ.eps')

figure  % plot of the translations
plot(N(10,:),'LineWidth',2)
hold on
xlim([-5,size(N,1)+5])
ylim([-7,17])
plot(N(ceil(size(N,1)/4),:),'LineWidth',2)
plot(N(3*ceil(size(N,1)/4),:),'LineWidth',2)
plot(N(ceil(size(N,1)/2),:),'LineWidth',2)
plot(N(end-10,:),'LineWidth',2)
grid on
print('-depsc','shifts.eps')
```

Simulation.m

```
close all
clear
clc

LxCGS = 160; %Domain size in cm
lCGS = 1; %length of the pressure field in cm
```

```
m = 2^10; %number of spatial intervals
TauCGS = .5; %Duration of surface forcing
TfCGS = TauCGS; %Length of simulation
Nt = 150; %number of temporal intervals
rhoCGS = 1; %density in CGS
sigmaCGS = 70; %Surface tension in CGS
gCGS = 980; %gravity in CGS

%%
%Units
L = lCGS; %Units of length
T = TauCGS; %Units of time
M = rhoCGS*L^3; %Units of mass

%Dimensionless groups
Fr = L/(T^2*gCGS);
We = rhoCGS*L^3/(T^2*sigmaCGS);

%Non-dimensionlising data
Lx = LxCGS/L; %Dimensionless length of the domain
Tf = TfCGS/T; %Dimensionless time of simulation

%Spatial discretisation
dx = Lx/m; %dimensionless width of the mesh intervals
xVec = (-Lx/2:dx:Lx/2)';

%DtN operator
line = zeros(1,m+1);
line(1) = 1+11/3;
line(2) = -16/9;
line(3) = -1/18;
for nn = 3:2:m
    an = -nn/(nn-1) +(nn+1/2)*log((nn+1)/(nn-1)) -1;
    bn =                -2*nn*log((nn+1)/(nn-1)) +4;
    cn = -nn/(nn+1) +(nn-1/2)*log((nn+1)/(nn-1)) -1;
    line(nn) = line(nn)+an;
    line(nn+1) = line(nn+1)+bn;
    line(nn+2) = line(nn+2)+cn;
end
line = line/(pi*dx);
N = zeros(m+1,m+1);
for ii = 1:m+1
    N(ii,:) = [fliplr(line(2:ii)),line(1:m-ii+2)];
end

%Temporal discretisation
dt = Tf/Nt;
tVec = 0:dt:Tf;
```

```
DXX = diag(-2*ones(1,m+1))+diag(ones(1,m),-1)+diag(ones(1,m),1)
    ;
A = [[eye(m+1),-dt*N/2];[ dt*eye(m+1)/(2*Fr)-dt*DXX/(2*We),eye(
    m+1)]];
B = [[eye(m+1), dt*N/2];[-dt*eye(m+1)/(2*Fr)+dt*DXX/(2*We),eye(
    m+1)]];

VecOld= zeros(2*m+2,1);

Ps = zeros(m+1,Nt+1);
for kk = 1:length(tVec)
    Ps(:,kk) = exp(-xVec.^2)'*(.5-.5*cos(2*pi*tVec(kk)));
end
fig1 = figure
surf(tVec,xVec,Ps,'LineStyle','none','FaceColor','interp')
print('-depsc','surf.eps')

%%
vidObj = VideoWriter('Pressed','MPEG-4');
open(vidObj);

fig2 = figure
fig2 = plot(xVec,VecOld(1:m+1),'LineWidth',2)
set(gca,'ylim',[-.02 .02],'xlim',[-Lx/2 Lx/2])
grid on
set(gca,'FontSize',16)
xlabel('$x/l$','interpreter','latex','FontSize',24)
ylabel('$\frac{\eta}{l}\␣\␣\␣\␣$','interpreter','latex','
    FontSize',24,'rotation',0)
title(['$t␣/␣\tau␣\,␣=␣\,$',num2str(tVec(1))],'interpreter','
    latex','FontSize',24)
for kk = 1:Nt
%     pause
    VecNew = A\(B*VecOld+[zeros(m+1,1);-dt*(Ps(:,kk)+Ps(:,kk+1)
        )]);
    fig2 = plot(xVec,VecNew(1:m+1),'LineWidth',2);
    set(gca,'ylim',[-.02 .02],'xlim',[-Lx/2 Lx/2])
    grid on
    set(gca,'FontSize',16)
    xlabel('$x/l$','interpreter','latex','FontSize',24)
    ylabel('$\frac{\eta}{l}\␣\␣\␣\␣$','interpreter','latex','
        FontSize',24,'rotation',0)
    title(['$t␣/␣\tau␣\,␣=␣\,$',num2str(tVec(kk+1))],'
        interpreter','latex','FontSize',24)
    if kk == 15
        print('-depsc','simul1.eps')
    elseif kk == 55
        print('-depsc','simul2.eps')
    elseif kk == 95
```

```
        print('-depsc','simul3.eps')
    elseif kk == 135
        print('-depsc','simul4.eps')
    end
    currFrame = getframe(gcf);
    writeVideo(vidObj,currFrame);
    VecOld = VecNew;
end
close(vidObj);
```

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